AE4ASM003 — Linear Modelling (incl. F.E.M.)

Homework Assignment 4 (Part 1)

AI assistance was used to format tables in LaTeX and improve presentation.

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Introduction

This part of the assignment focuses on the formulation of a two–dimensional triangular membrane element with four nodes. The element shown in Figure 1 is analysed in a local Cartesian coordinate system. The objective is to derive the corresponding shape functions and later use them to formulate the element's strain–displacement matrix and equivalent nodal forces.

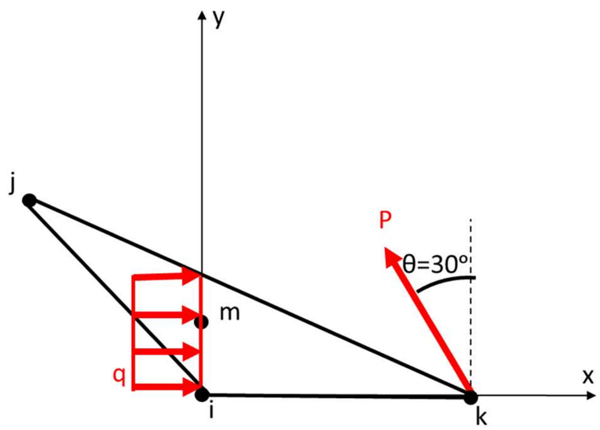

Figure 1: Geometry of the triangular membrane element with one internal node.

$$i(0,0), \qquad j\left(-\tfrac{w}{3},\, h\right), \qquad k(w,0), \qquad m(0,\tfrac{h}{2})$$

Question a)

The in-plane displacements of the element are defined separately in the x- and y-directions as

$$\begin{cases} u^{(e)} = c_1 + c_2 x + c_3 y + c_4 xy, \\ v^{(e)} = c_5 + c_6 x + c_7 y + c_8 xy, \end{cases}$$

where $u^{(e)}$ and $v^{(e)}$ represent the displacement components of the element in the x- and y-directions, respectively. Because the mathematical form of $v^{(e)}$ is analogous to that of $u^{(e)}$, the following derivation is performed only for the x-direction displacement field.

The displacement field is therefore assumed as

$$u(x,y) = c_1 + c_2 x + c_3 y + c_4 xy,$$

where c_1, c_2, c_3, c_4 are unknown constants to be determined from the nodal displacements.

At each node, the displacement is obtained by substituting the corresponding coordinates:

$$\begin{aligned} u_i &= u(0,0) = c_1, \\ u_j &= u(-\tfrac{w}{3}, h) = c_1 - \tfrac{w}{3}c_2 + hc_3 - \tfrac{wh}{3}c_4, \\ u_k &= u(w,0) = c_1 + c_2 w, \\ u_m &= u(0, \tfrac{h}{2}) = c_1 + c_3 \tfrac{h}{2}. \end{aligned}$$

From the first, third and fourth equations, the constants c_1, c_2, c_3 can be expressed in terms of the nodal displacements as

$$c_1 = u_i, \qquad c_2 = \frac{u_k - u_i}{w}, \qquad c_3 = \frac{2(u_m - u_i)}{h}.$$

Substituting these expressions into the second equation gives

$$u_j = u_i - \tfrac{1}{3}(u_k - u_i) + 2(u_m - u_i) - \tfrac{wh}{3}c_4,$$

from which the remaining coefficient is obtained:

$$c_4 = \frac{-2u_i - u_k + 6u_m - 3u_j}{wh}.$$

Substituting all four coefficients into the original displacement field yields

$$u(x,y) = u_i + \Big(\frac{u_k - u_i}{w}\Big)x + \Big(\frac{2(u_m - u_i)}{h}\Big)y + \Big(\frac{-2u_i - u_k + 6u_m - 3u_j}{wh}\Big)xy.$$

Regrouping the terms according to the nodal displacements gives

$$u(x,y) = N_i(x,y)u_i + N_j(x,y)u_j + N_k(x,y)u_k + N_m(x,y)u_m,$$

where the shape functions are identified as

$$\begin{aligned}
N_i(x,y) &= 1 - \frac{x}{w} - \frac{2y}{h} - \frac{2xy}{wh},\\
N_j(x,y) &= -\frac{3xy}{wh},\\
N_k(x,y) &= \frac{x}{w} - \frac{xy}{wh},\\
N_m(x,y) &= \frac{2y}{h} + \frac{6xy}{wh}.
\end{aligned}$$

These shape functions satisfy all standard interpolation properties. At their own nodes they equal unity and vanish at all others:

$$N_i(0,0) = 1, \quad N_j(-\tfrac{w}{3}, h) = 1, \quad N_k(w,0) = 1, \quad N_m(0, \tfrac{h}{2}) = 1,$$

and $N_i + N_j + N_k + N_m = 1$ everywhere inside the element. They span the polynomial space $\{1, x, y, xy\}$, matching the assumed displacement field.

Additionally, the displacement vector in the x- and y-directions can be written compactly as

$$\begin{bmatrix} u(x,y) \\ v(x,y) \end{bmatrix} = \underbrace{\begin{bmatrix} N_i(x,y) & 0 & N_j(x,y) & 0 & N_k(x,y) & 0 & N_m(x,y) & 0 \\ 0 & N_i(x,y) & 0 & N_j(x,y) & 0 & N_k(x,y) & 0 & N_m(x,y) \end{bmatrix}}_{\mathbf{N}(x,y)} \begin{bmatrix} u_{ix} \\ u_{iy} \\ u_{jx} \\ u_{jy} \\ u_{kx} \\ u_{ky} \\ u_{mx} \\ u_{my} \end{bmatrix}.$$

Question b)

For this triangular membrane element, only in-plane displacements are considered. The element is assumed to be in a plane stress state, meaning

$$\sigma_z = \tau_{yz} = \tau_{xz} = 0.$$

Hence, all stresses and strains are confined to the element's mid-plane.

The membrane strain vector is defined as

$$\boldsymbol{\varepsilon}(x, y) = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \mathbf{B}(x, y) \begin{bmatrix} u_{ix} \\ u_{iy} \\ u_{jx} \\ u_{jy} \\ u_{kx} \\ u_{ky} \\ u_{mx} \\ u_{my} \end{bmatrix}.$$

Since the displacements at the nodes are constant, only the partial derivatives of the shape functions are accounted for in the strain-displacement matrix. The partial derivatives of the shape functions are

$$\begin{aligned} \frac{\partial N_i}{\partial x} &= -\frac{1}{w} - \frac{2y}{wh}, & \frac{\partial N_i}{\partial y} &= -\frac{2}{h} - \frac{2x}{wh}, \\ \frac{\partial N_j}{\partial x} &= -\frac{3y}{wh}, & \frac{\partial N_j}{\partial y} &= -\frac{3x}{wh}, \\ \frac{\partial N_k}{\partial x} &= \frac{1}{w} - \frac{y}{wh}, & \frac{\partial N_k}{\partial y} &= -\frac{x}{wh}, \\ \frac{\partial N_m}{\partial x} &= \frac{6y}{wh}, & \frac{\partial N_m}{\partial y} &= \frac{2}{h} + \frac{6x}{wh}. \end{aligned}$$

Thus,the strain-displacement matrix is defined as

$$\mathbf{B}(x, y) = \begin{bmatrix} \dfrac{\partial N_i}{\partial x} & 0 & \dfrac{\partial N_j}{\partial x} & 0 & \dfrac{\partial N_k}{\partial x} & 0 & \dfrac{\partial N_m}{\partial x} & 0 \\ 0 & \dfrac{\partial N_i}{\partial y} & 0 & \dfrac{\partial N_j}{\partial y} & 0 & \dfrac{\partial N_k}{\partial y} & 0 & \dfrac{\partial N_m}{\partial y} \\ \dfrac{\partial N_i}{\partial y} & \dfrac{\partial N_i}{\partial x} & \dfrac{\partial N_j}{\partial y} & \dfrac{\partial N_j}{\partial x} & \dfrac{\partial N_k}{\partial y} & \dfrac{\partial N_k}{\partial x} & \dfrac{\partial N_m}{\partial y} & \dfrac{\partial N_m}{\partial x} \end{bmatrix}.$$

Substituting the explicit derivatives yields

$$\mathbf{B}(x, y) = \begin{bmatrix} -\dfrac{1}{w} - \dfrac{2y}{wh} & 0 & -\dfrac{3y}{wh} & 0 & \dfrac{1}{w} - \dfrac{y}{wh} & 0 & \dfrac{6y}{wh} & 0 \\ 0 & -\dfrac{2}{h} - \dfrac{2x}{wh} & 0 & -\dfrac{3x}{wh} & 0 & -\dfrac{x}{wh} & 0 & \dfrac{2}{h} + \dfrac{6x}{wh} \\ -\dfrac{2}{h} - \dfrac{2x}{wh} & -\dfrac{1}{w} - \dfrac{2y}{wh} & -\dfrac{3x}{wh} & -\dfrac{3y}{wh} & -\dfrac{x}{wh} & \dfrac{1}{w} - \dfrac{y}{wh} & \dfrac{2}{h} + \dfrac{6x}{wh} & \dfrac{6y}{wh} \end{bmatrix}.$$

Question c)

The present four-node triangular membrane element represents an enhanced formulation compared with the three-node constant strain triangle (CST) introduced in class. Its displacement field is expressed with an additional cross term xy, which introduces curvature into the deformation pattern. As a result, the strain components are no longer constant within the element but vary linearly across its area. This enables the element to reproduce any linearly varying stress field exactly and therefore to satisfy the linear patch test. In practical terms, this improvement allows the element to represent complex deformation behaviour, such as gradual strain variations or bending, induced stress gradients—within a single element, rather than requiring multiple CSTs to achieve the same result. Consequently, the element provides a smoother and more realistic stress distribution and improves the convergence rate of the analysis.

The higher-order interpolation also improves the prediction of stress concentrations and bending-induced variations that a CST element cannot capture. When a structure experiences non-uniform loading or local bending, the stress and strain fields vary linearly within the plane. The inclusion of the xy term allows the element to capture these gradients accurately, enhancing its ability to represent plate- or membrane-like behaviour with in-plane curvature. This richer displacement field means that, for a given mesh density, the modified element yields higher numerical accuracy and smoother strain transitions; alternatively, the same level of accuracy can be achieved with fewer elements.

Retaining the triangular geometry provides an important modelling advantage: such elements can fill arbitrary or irregular domains without producing gaps or overlapping regions. This makes them especially convenient for automated mesh generation and for regions with curved or tapering boundaries, where quadrilateral meshes are harder to create. Furthermore, the triangular shape remains computationally efficient and adaptable to complex geometries, especially in aerospace applications involving fuselage skins or wing panels where geometric irregularities are common.

In contrast to a four-node bilinear quadrilateral element, the triangular element offers greater meshing freedom and better tolerance of irregular boundaries. However, quadrilateral elements typically deliver slightly higher numerical accuracy for the same number of degrees of freedom when the geometry is regular and the stress field is smooth. This is because quadrilaterals use isoparametric bilinear shape functions, where both geometry and displacement are interpolated in the same natural coordinate system (ξ, η). The presence of the cross term $\xi\eta$ enables bilinear mapping, allowing curved edges and two-directional deformation to be represented accurately. As a result, quadrilateral meshes produce smoother stress and strain distributions and can model curved or twisted surfaces with fewer elements. Nonetheless, this comes at the cost of a more complex formulation and the need for numerical integration through the Jacobian matrix, which increases computational cost and can introduce numerical instability if elements are highly distorted.

Triangular elements, on the other hand, remain more robust and flexible for irregular meshes but can be somewhat more sensitive to geometric distortion when poorly shaped (e.g., with sharp interior angles or high aspect ratios). Excessive distortion may degrade stiffness matrix conditioning and increase numerical error, though this effect is mitigated by the improved displacement interpolation introduced by the xy term.

In summary, the four-node triangular membrane element combines the simplicity and flexibility of triangular meshing with the improved strain representation of higher-order interpolation. Although not as efficient as a well-shaped quadrilateral element on structured meshes, it provides a balanced compromise between computational efficiency, accuracy, and geometric adaptability, offering a substantial improvement over the CST formulation for capturing linearly varying strain and stress fields in practical engineering applications.

Question d)

As observed in Figure 1, both a point and a distributed load are applied to the element. The point load P is applied at node k and can be decomposed into two components:

$$F_{kx} = -P\sin(30^\circ) = -\tfrac{1}{2}P, \qquad F_{ky} = P\cos(30^\circ) = \tfrac{\sqrt{3}}{2}P.$$

The distributed load acts only along the $x = 0$ edge, in the x-direction, with intensity q. It can be expressed through a 2×1 matrix:

$$Q = \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} q \\ 0 \end{pmatrix}.$$

The external work done on the element can be expressed as:

$$W^{(e)} = \int_A (U)^T(P)\,dA = u_{kx}F_{kx} + u_{ky}F_{ky} + \int_A [N]^T(U)^T(Q)\,dA.$$

Thus, the force vector can be defined as:

$$F^{(e)} = \frac{\partial W^{(e)}}{\partial U} = F_{kx} + F_{ky} + \int_A [N]^T(Q)\,dA.$$

Solving the integral in the previous expression:

$$\int_A [N]^T(Q)\,dA = \int_A \begin{bmatrix} N_i & 0 \\ 0 & N_i \\ N_j & 0 \\ 0 & N_j \\ N_k & 0 \\ 0 & N_k \\ N_m & 0 \\ 0 & N_m \end{bmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} dA = \int_A \begin{bmatrix} N_i & 0 \\ 0 & N_i \\ N_j & 0 \\ 0 & N_j \\ N_k & 0 \\ 0 & N_k \\ N_m & 0 \\ 0 & N_m \end{bmatrix} \begin{pmatrix} q \\ 0 \end{pmatrix} dA.$$

Since the distributed load acts along $x = 0$, the area integral can be replaced by a line integral from $y = 0$ to $y = \frac{3}{4}h$. Along this edge, $N_i = 1 - \frac{2y}{h}$, $N_m = \frac{2y}{h}$, and $N_j = N_k = 0$. Hence:

$$\int_A [N]^T(Q)\,dA = qt\int_0^{3h/4} \begin{bmatrix} N_i \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ N_m \\ 0 \end{bmatrix} dy = qt \begin{bmatrix} \int_0^{3h/4} \left(1 - \frac{2y}{h}\right) dy \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \int_0^{3h/4} \left(\frac{2y}{h}\right) dy \\ 0 \end{bmatrix} = qt \begin{bmatrix} \frac{3}{16}h \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{9}{16}h \\ 0 \end{bmatrix}.$$

Finally, by adding the point load components F_{kx} and F_{ky}, the total element force vector is obtained:

$$F^{(e)} = \begin{bmatrix} \frac{3}{16}qth \\ 0 \\ 0 \\ 0 \\ -\frac{1}{2}P \\ \frac{\sqrt{3}}{2}P \\ \frac{9}{16}qth \\ 0 \end{bmatrix}.$$

References

[1] D. Peeters, *Linear Modelling (incl. FEM)* , TU Delft, Version 4, 2025.

AE4ASM003 — LINEAR MODELLING (INCL. F.E.M.)

Homework Assignment 4 (Part 2)

AI assistance was used to format tables in LaTeX and improve presentation.

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Introduction

A 400×700 mm panel is tested in in-plane compression. The testing machine records a total compressive load of $P = 10$ kN. During the experiment, the top and bottom 50 mm of the specimen are embedded in epoxy.

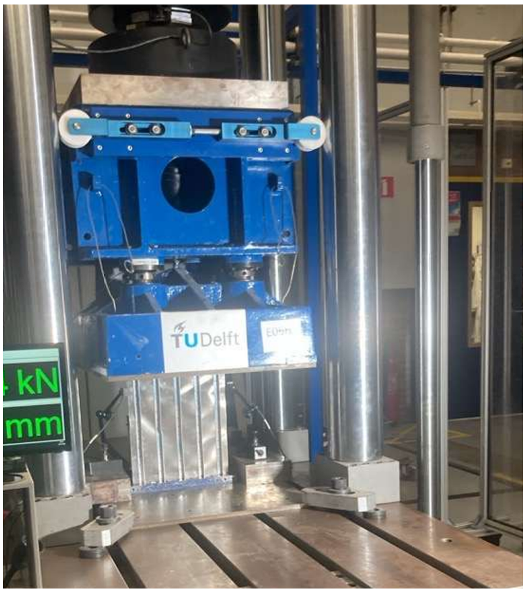

Figure 1: Experimental compression setup. The epoxy grips ensure uniform load transfer and prevent slip or rotation of the panel edges.

Question a)

The test is representative of the upper part of the wingbox in cruise because the aluminium panel is loaded in uniform in-plane compression, reproducing the dominant stress state in the upper wing skin during steady, level flight. In cruise, the aerodynamic lift acting on the wing generates an upward bending moment along the span. As a result, the upper surface of the thin-walled wingbox shortens and carries axial compressive membrane stresses, while the lower surface elongates in tension. In the experiment, the panel is compressed between two epoxy-embedded ends that restrain translation and rotation, effectively simulating the constraint provided by spars and ribs, which hold the skin in place and transfer the compressive load through the box. Thus, the test reproduces the correct type of loading (in-plane compression) and a reasonable approximation of edge restraint, making it a valid local representation of the upper wingbox skin's behaviour in cruise.

However, it is not fully representative because, in a real aircraft during cruise, the compressive stress in the upper wing skin arises from global bending of the entire wingbox under aerodynamic lift, whereas in the laboratory test the load is applied directly and uniformly through rigid end platens. This difference means that the stress distribution and boundary stiffness in the test are not the same as those in the actual structure. In the real wingbox, the compressive load is introduced gradually through the surrounding structure, via spars, ribs, and the lower skin in tension, producing non-uniform bending-induced stresses and allowing limited edge rotation and shear coupling. In contrast, the test imposes pure, uniform compression with fully clamped ends, generating a more constrained state and eliminating the natural load redistribution that occurs in flight. Consequently, while the test accurately captures the magnitude and nature of local compression in the upper skin, it does not reproduce the way that compression is generated and transmitted within the wingbox during cruise.

Question b)

The experimental setup (see Fig. 1) consists of an aluminium panel compressed between two epoxy gripping blocks bonded at its top and bottom edges. In the physical test, the epoxy distributes the applied load uniformly along the panel width and prevents local crushing of the aluminium. Because the epoxy geometry and mechanical properties are not provided, it is not modelled explicitly. Instead, its effect is represented through boundary conditions that mimic the constraint it imposes on the panel surfaces.

The bottom epoxy grip holds the specimen fixed to the testing-machine base; therefore, the bottom edge of the panel is modelled as encastre, all translational and rotational degrees of freedom are set to zero. This boundary condition reproduces the way the epoxy prevents any movement of the lower edge during loading.

The top epoxy grip transfers the compressive load from the machine's actuator. In the model, the same load is represented as a uniform compressive edge load of 25 N/mm applied along the top edge of the panel, directed downward. This value corresponds to the total applied force of 10 kN divided by the panel width of 400 mm, ensuring that the load is distributed evenly, just as the epoxy and steel platen do in the test.

The top epoxy grip transfers the compressive load from the machine's actuator. In the model, the same load is represented as a uniform compressive edge load of 25 N/mm applied along the top edge of the panel, directed downward. This value corresponds to the total applied force of 10 kN divided by the panel width of 400 mm, ensuring that the load is evenly distributed, just as the epoxy and steel platen do in the physical test. The epoxy at the top allows vertical movement while constraining lateral and rotational degrees of freedom, which is implemented through the boundary conditions $U_1 = U_3 = UR_1 = UR_2 = UR_3 = 0, U_2$ free. This configuration ensures that the load is transmitted purely in compression while preventing rigid-body motion or local tilting at the upper edge.

To turn this panel into a finite-element model, the problem was idealised as a three-dimensional shell. This type of part was chosen because it allows the panel to be represented as a thin-walled structure, where the thickness is much smaller than the in-plane dimensions. The panel was modelled using S4R shell elements, which are suitable for thin or moderately thick plates subjected to membrane and bending effects. These elements employ reduced integration, making them computationally efficient while accurately capturing both the in-plane stress and strain and any small out-of-plane deformation that may develop under compression. Since bending stiffness in this configuration is limited and the response is dominated by membrane action, the S4R formulation provides a reliable representation of the panel behaviour.

The panel itself is idealised as a 400×600 mm flat, homogeneous aluminium shell with a thickness of 1.5 mm. The mesh size of approximately 20×20 mm provides a good balance between accuracy and computational efficiency for capturing the global membrane behaviour.

It is important to note that, throughout the development of this report, the effect of the symmetry of the panel could be used as a way to decrease the computational effort of the model. Instead of modelling the full 400 mm width, it would have been possible to model only half (200 mm) and apply a symmetry boundary condition on the cut side, restricting displacement in the x-direction ($U_1 = 0$).

The same loading and boundary conditions would then be applied to this half-model, producing the same results as the full panel (except in the later stiffened configuration of Question 2.4). Nevertheless, the entire panel was modelled here for consistency and direct comparison with the physical test.

A view of the final finite-element model with the applied boundary conditions and loads is shown in Fig. 2.

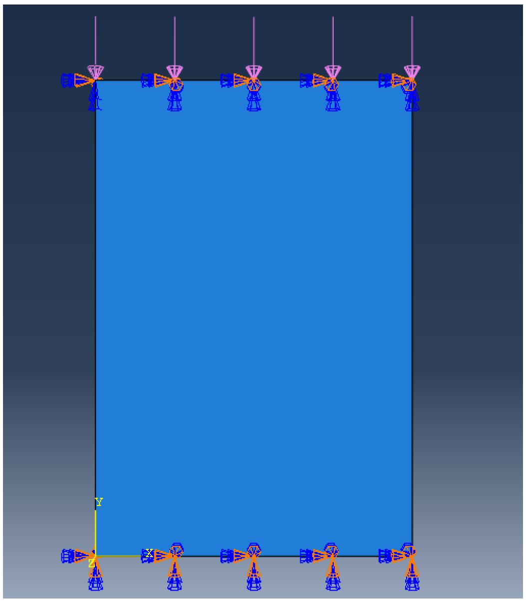

Figure 2: Finite-element model of the aluminium panel with boundary conditions and compressive edge load.

Question c)

The obtained displacement and stress fields for the flat aluminium panel under compression are presented in Figures 3 and 4. The overall behaviour corresponds to a linear elastic pre-buckling response, as expected for the applied compressive load of 10 kN.

To verify the numerical results, an analytical estimation can be made. The applied load corresponds to a uniform compressive line load of $q = 25$ N/mm, which produces an average axial stress of

$$\sigma_y = \frac{-F}{A} = \frac{-10{,}000}{1.5 \times 400} \approx -16.7 \text{ MPa}.$$

Assuming linear elasticity with $E = 70$ GPa, the resulting strain and vertical shortening are estimated as

$$\varepsilon_y = \frac{\sigma_y}{E} \approx -2.4 \times 10^{-4}, \qquad \Delta L = \varepsilon_y L = -2.4 \times 10^{-4} \times 600 \approx -0.14 \text{ mm}.$$

This analytical estimation provides a reliable reference for assessing the numerical solution.

Displacement analysis

The displacements in both the x and y directions are symmetric (in magnitude) about the vertical mid-plane of the panel. This behaviour is expected due to the symmetry of both the geometry and the applied boundary conditions. Moreover, the boundary conditions are correctly satisfied: along the top edge, the imposed constraints $U_1 = U_3 = 0$ ensure that no lateral or out-of-plane displacements occur, while at the bottom edge, the encastre condition is respected, as the vertical displacements are approximately zero for the bottom nodes. The deformation pattern confirms that the imposed constraints and loading were properly implemented and that the response is physically consistent.

The in-plane displacement U_1 is antisymmetric about the mid-plane: the right side moves slightly towards $+x$ and the left towards $-x$, reflecting that as the panel is compressed in the y-direction, it tends to expand transversely (Poisson effect). The maximum value is 0.01436 mm and is located at the edge of the panel's side.

The vertical displacement U_2 field shows a smooth, nearly uniform shortening along the height of the panel, reaching a maximum value of approximately 0.14 mm at the top edge, fully consistent with the analytical prediction.

The out-of-plane displacement U_3 remains negligible, confirming that no buckling or secondary bending occurs and that the model behaves as a purely planar, linear-elastic structure.

As expected, the panel is contracting in the y-direction and expanding on both sides in the x-direction, due to the the clamped bottom and distributed tensile load.

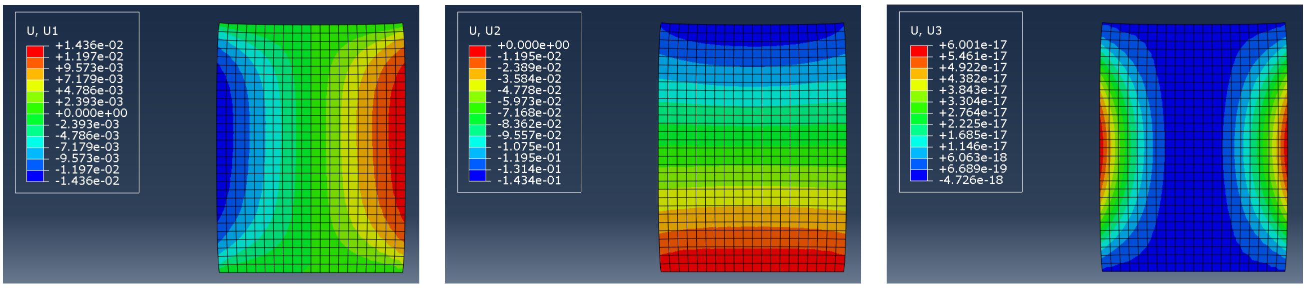

Figure 3: Displacement fields for the flat panel: (left) U_1, (centre) U_2, and (right) U_3.

Stress analysis

In terms of stress distribution, all stress components (S_{11}, S_{22}, and S_{12}) are symmetric in magnitude about the vertical axis, as expected.

The transverse stress S_{11}, acting in the x-direction, shows tensile values across most of the central region of the panel, indicating lateral expansion. Near the epoxy grips, however, S_{11} becomes compressive, forming narrow bands of negative stress at the upper and lower edges. These regions appear because the boundary conditions restrict lateral expansion, forcing the material near the constrained edges to experience local compression in the x-direction.

The S_{22} component dominates the response and represents the applied compressive load acting vertically on the panel. The stress distribution shows a clear gradient along the height: the largest compressive stresses occur near the bottom corners, while smaller compressive values appear towards the top. This behaviour is consistent with the boundary conditions imposed on U_2. The bottom edge, which is encastre and fully restrained in the vertical direction, must react the entire applied load; as a result, compressive stresses accumulate near this region where vertical deformation is prevented. Moving upward, the stresses progressively decrease in magnitude because the top edge is free to move axially (U_2 free) and therefore experiences less constraint-induced amplification. In the central part of the panel, away from the boundaries, the compressive field remains nearly uniform with an average value of approximately -17 MPa, in excellent agreement with the analytical estimation of the nominal stress. Slight stress concentrations persist near the grips, where the boundary conditions restrict lateral contraction, but overall the contour pattern confirms that the load transfer and constraint effects are physically consistent with the intended membrane compression state.

The shear component S_{12} remains negligible throughout most of the panel, except at the corners, where shear peaks (up to ± 3.3 MPa) appear due to the constraint transition between free and fixed zones.

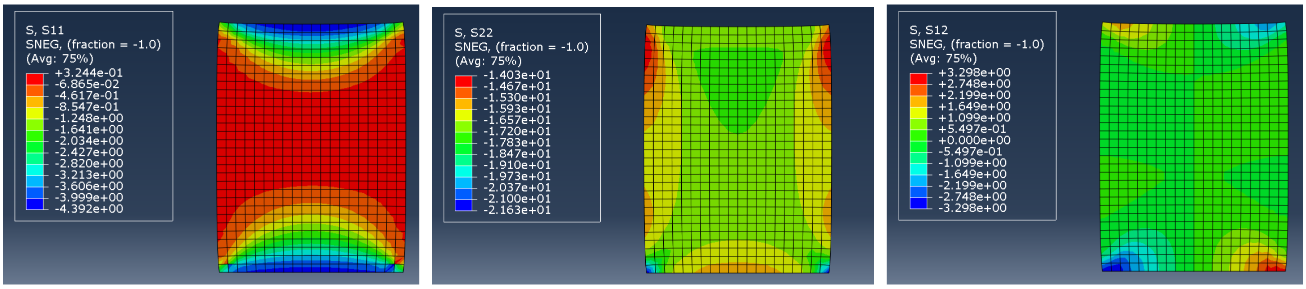

Figure 4: Stress fields for the flat panel: (left) S_{11}, (centre) S_{22}, and (right) S_{12}.

Effect of idealisation and model evaluation

The small deviations between the analytical and numerical stress fields arise mainly from the modelling idealisation of the epoxy grips. Representing the epoxy regions as fixed boundaries restricts lateral contraction and rotation, which locally increases stiffness near the edges and produces slight stress concentrations and small shear values at the corners. These effects are numerical artefacts of the simplified boundary modelling and do not represent physical behaviour.

Away from the constrained edges, the panel response is uniform and dominated by axial compression, confirming that the chosen boundary conditions successfully reproduce the intended pre-buckling membrane behaviour. The analysis is linear static, assuming small deformations and rotations, which is appropriate for the load level considered.

Ideally, the epoxy layers could have been modelled explicitly to represent their finite stiffness and shear transfer, allowing for a more realistic deformation and stress distribution near the grips. However, this was not implemented due to the lack of detailed material properties and the scope of the assignment. Similarly, it would have been useful to perform an eigenvalue buckling analysis to predict the critical load and verify the onset of instability, but this was beyond the current modelling requirements. Despite these simplifications, the results obtained are physically meaningful and fully reliable within the assumptions of the linear pre-buckling regime.

Question d)

In order to reduce the axial shortening of the panel under compression, two design modifications were evaluated: (I) increasing the thickness of a central strip and (II) adding two longitudinal L-shaped stiffeners modelled as 1D beams.

Option I

The panel is idealised as a shell, with a central 100 mm band over the 600 mm gauge length thickened symmetrically about the mid-surface. The thickness therefore changes from $t = 1.5$ mm to $t = 3.5$ mm

(+1 mm on each side), ensuring that no artificial membrane-bending coupling is introduced. The same boundary conditions and loading as in Question c) were applied.

This modification increases the effective axial area of the panel from $\mathrm{A}_0 = 1.5 \times 400 = 600\ \mathrm{mm}^2$ $A_1 = 3.5 \times 100 + 1.5 \times 300 = 800\ \mathrm{mm}^2$, corresponding to an approximate 33% increase in cross-sectional area. Because the axial strain in the pre-buckling regime is inversely proportional to the axial stiffness, the theoretical shortening can be estimated as

$$\Delta L_1 \approx \Delta L_0 \frac{A_0}{A_1} = 0.14\ \mathrm{mm} \times \frac{600}{800} \approx 0.105\ \mathrm{mm},$$

which represents a predicted reduction of about 25% compared to the baseline configuration.

The numerical results confirm this expected behaviour. The vertical displacement field U_2 shows a smooth and symmetric contraction along the height of the panel, with a maximum value of approximately -0.1126 mm at the top edge, which corresponds to an actual reduction of 21.47% relative to the baseline -0.1434 mm. The difference from the analytical estimate is minor and is attributed to the local constraint and shear transfer at the epoxy grips. The top edge, which was previously straight in the baseline configuration, now exhibits a slightly convex shape, reflecting the higher stiffness of the central region that restrains local shortening. The displacement U_2 is higher (in magnitude) at the thinner sides and lower at the thicker central band, resulting in a small curvature difference along the width of the top edge. This curvature is small and remains within the elastic, pre-buckling range, but it reveals that the thickened region redistributes the deformation more uniformly along the panel height.

The horizontal displacement U_1 maintains the same antisymmetric pattern observed in the baseline model, indicating that the Poisson-driven lateral movement is not significantly affected by the added thickness. The maximum value decreased but remained around $\pm 1.0 \times 10^{-2}$ mm, confirming that the modification primarily increases axial stiffness without notably altering lateral behaviour. The out-of-plane displacement U_3 remains negligible (order of 10^{-17} mm), demonstrating that the panel continues to behave as a planar structure, unaffected by bending or secondary instabilities.

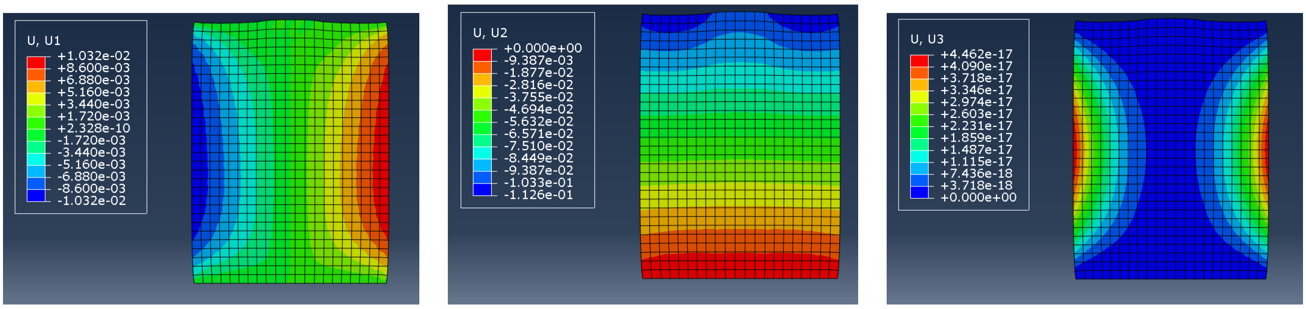

Figure 5: Option I (thickened strip): displacement fields U_1 (left), U_2 (centre), and U_3 (right).

Overall, Option I provides a clear improvement in vertical stiffness, leading to a measurable reduction in axial shortening without introducing parasitic bending or instability. The modification maintains the desired symmetry and results in a uniform, stable pre-buckling response. However, the lateral deformation pattern (U_1) remains essentially unchanged, indicating that the increased stiffness mainly benefits the vertical direction, with limited influence on the transverse contraction.

In practical terms, manufacturing such a panel with a variable thickness poses notable challenges. The most accurate method would involve starting with a uniform aluminium sheet of 3.5 mm thickness and milling down the side regions to 2 mm, thereby producing the desired 3.5 mm central band. Although precise, this machining process is expensive, time-consuming, and leads to considerable material waste, especially at larger production scales. An alternative approach would be to adhesively bond two thin 1 mm plates to the central region of a 1.5 mm sheet. While simpler to manufacture, this configuration would reduce mechanical performance due to the limited shear strength of adhesive joints and the risk of delamination under compressive loading. Therefore, although the thickened-strip solution offers an effective means of reducing axial shortening in simulation, its practical implementation would require careful consideration of manufacturing feasibility and joint integrity in real-world applications.

Option II

In this configuration, two longitudinal stiffeners with L-shaped cross-sections were introduced, extending over the full height of the panel. The stiffeners were modelled using 1D beam elements and tied to the mid-surface of the shell along their central lines to ensure compatible displacements between the shell and the beams. Both stiffeners had the same orientation and geometry, defined according to the provided dimensions. The aluminium material properties were assumed identical to those of the panel. The same boundary conditions and loading as in Question c) were applied.

Contrary to the baseline model, the overall displacement field is not fully symmetric about the vertical mid-plane. This asymmetry is mainly attributed to the geometry and orientation of the L-shaped stiffeners. The L-section lacks symmetry about both principal axes, and since both stiffeners are oriented in the same direction, one side of the panel experiences a slightly different stiffness and deformation response. The asymmetry could be mitigated by rotating one of the stiffeners so that the assembly becomes geometrically balanced.

The addition of the stiffeners increases the effective axial stiffness of the panel by introducing longitudinal bending rigidity and an additional load-carrying path. The numerical results confirm this expected behaviour. The vertical displacement field U_2 shows a generally smooth contraction along the panel height, with a maximum value of approximately -0.1127 mm at the top edge. This corresponds to an improvement of about 21.40% compared to the baseline configuration (-0.1434 mm), confirming that the stiffeners effectively reduce the global axial shortening. The U_2 field remains mostly symmetric, but small local variations appear along the stiffener lines, where the beams attract load and restrain the shell deformation. The top edge becomes slightly non-planar, with the region between the stiffeners shortening more than their immediate surroundings, creating a mild convex curvature.

The horizontal displacement field U_1 retains the general antisymmetric Poisson-driven pattern but now shows noticeable differences between the two sides of the panel. The maximum magnitude reaches approximately $\pm 1.1 \times 10^{-2}$ mm, slightly higher on one side due to the unequal stiffness contribution of the stiffeners. The out-of-plane displacement field U_3 exhibits a clear bulging pattern, with a maximum amplitude of about $+2.8 \times 10^{-2}$ mm. This deformation arises from the eccentric positioning of the stiffeners relative to the shell's mid-surface, which induces local bending–membrane coupling even under a purely axial compressive load. The curvature remains moderate and elastic, indicating that the system still behaves within the pre-buckling range.

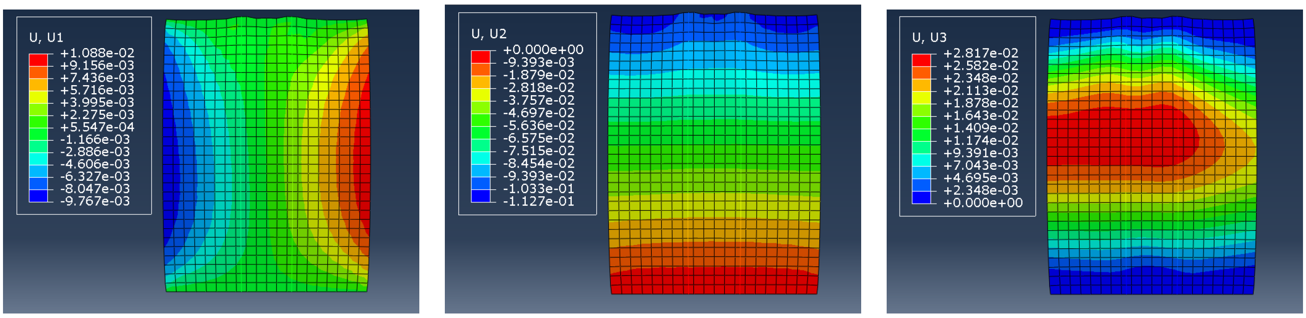

Figure 6: Option II (two L-shaped stiffeners): displacement fields U_1 (left), U_2 (centre), and U_3 (right).

Overall, the stiffened configuration effectively reduces the vertical displacement U_2, demonstrating the expected increase in axial stiffness. However, the eccentricity of the stiffeners introduces local bending and slight asymmetry in the deformation field. All displacements remain within the linear-elastic pre-buckling regime, validating the physical consistency of the model.

From a practical perspective, this configuration is easier to manufacture than a variable-thickness plate. The use of standard L-section extrusions enables straightforward attachment to the base sheet by welding, riveting, or adhesive bonding. Nevertheless, such joints may act as local discontinuities, concentrating stresses and potentially reducing fatigue life. Moreover, careful assembly is required to ensure proper alignment and minimise additional eccentricities that could amplify bending effects.

Conclusion

Both design modifications achieve a comparable reduction in axial shortening, decreasing the maximum vertical displacement U_2 by about 21% relative to the baseline. Since the total volume of material in both configurations is 480,000 mm^3, their weight are identical, meaning this factor does not influence the comparison.

Although the overall U_2 values are nearly identical for both configurations, the thickened-strip design (Option I) provides slightly lower maximum horizontal displacements U_1, indicating marginally better control of the lateral contraction. More importantly, the out-of-plane displacement U_3 remains practically zero in this configuration, whereas the stiffened panel (Option II) develops a noticeable bulging effect. Therefore, Option I delivers the most balanced and mechanically desirable response, maintaining full symmetry, avoiding bending–membrane coupling, and minimising both vertical and lateral deformations.

From a practical standpoint, however, Option II remains simpler and cheaper to manufacture, as it can be built using standard L-section stiffeners and conventional joining techniques. Option I would require variable-thickness machining or precise bonding, increasing cost and complexity.

In summary, Option I provides the best overall structural performance, while Option II offers a more practical and economical solution. The optimal choice ultimately depends on whether the design prioritises mechanical precision or manufacturing efficiency.

Question e)

In this new configuration, the panel is subjected to tension instead of compression. The top part of the specimen is first clamped by the epoxy and the testing fixture, while the bottom part is loaded by the action of many small weights that are evenly distributed along its entire width. The epoxy remains present at both ends and continues to ensure a uniform transfer of load while restricting lateral and rotational movement of the gripped regions.

To idealise this configuration in the finite element model, the overall geometry, material properties, mesh density, and element formulation remain unchanged. The only modifications required concern the boundary conditions and the loading direction, so that the model correctly represents the new experimental procedure.

The top edge of the panel is modelled as encastre, reproducing the mechanical behaviour of the upper epoxy grip and fixture that hold the specimen fixed before loading. This boundary condition prevents any displacement or rotation along the top boundary (encastre) and can be written as $U_1 = U_2 = U_3 = UR_1 = UR_2 = UR_3 = 0$. This represents the “first clamped” stage of the test, where the upper epoxy block fixes the specimen completely.

The bottom edge, where the weights are attached through the epoxy layer, is free to move only in the vertical direction. The epoxy prevents lateral slip, out-of-plane deformation, and rotation, but it allows axial extension as the panel elongates under tension. This behaviour is represented by constraining five degrees of freedom while keeping the vertical displacement free: $U_1 = U_3 = UR_1 = UR_2 = UR_3 = 0, U_2$ free. This condition accurately reflects the behaviour of the lower epoxy layer, which guides the vertical motion of the edge while transmitting the tensile load produced by the suspended weights.

The load is introduced as a uniformly distributed vertical line load acting downward along the bottom edge. The total applied force corresponds to the combined weight of all suspended masses, $F = M_{\text{tot}}g$, and the equivalent uniform line load applied over the 400 mm width of the specimen is $q = \frac{F}{400\ \text{mm}}$ [N/mm]. This uniform load has the same mechanical effect as the presence of many small, equally spaced weights acting over the full width of the panel.

As in the compressive case, the setup remains symmetric with respect to the vertical mid-plane of the specimen. It would therefore be possible to model only half of the width by imposing a symmetry boundary condition along the centreline ($U_1 = 0$). Nevertheless, the complete model may be used for direct comparison with the experimental configuration and to avoid potential loss of detail in the stress and deformation fields.

References

[1] D. Peeters, *Linear Modelling (incl. FEM)* , TU Delft, Version 4, 2025.